

Marcus Oates

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Profile

- First employee and backend engineer at Voxworks, a Voice AI startup, where I led a ground-up restructure of the voice agent backend, its core product, and scaled call concurrency by 400%.
- Creator of Gym Junkie, a fitness tracking app now live on the Google Play Store and Apple App Store.
- Automation software engineer at Downer, building automated microservices with ML/AI, managing Azure resources and spearheading IaC adoption (GitHub Actions).
- Bachelor of Engineering (First Class Honours) / Computer Science (AI), UNSW: WAM 84.
- Short-term consultancy internship at Incat Crowther in naval engineering and architecture (2023).

Experience

Senior Software Engineer: Voxworks (Voice AI Startup)

Jan 2026 – Present

- As the startup's first employee, took operational control of the voice agent's backend architecture and led a ground-up restructure of the codebase, the company's core product, owning its reliability and ongoing development in production.
- Re-architected the core conversational logic from an accumulation of disjoint systems into a single deterministic flow, retiring substantial technical debt while maintaining interoperability with the live system so clients were served without interruption.
- Instrumented every agent decision so any call can be traced back to the exact state that produced it, making AI-assisted debugging the primary diagnostic path and cutting investigation time on conversational faults.
- Migrated the voice agent off LiveKit onto a fully self-hosted AWS telephony stack, gaining full control of SIP connections and call handoffs, increasing call concurrency by 400% and removing a fixed latency cost, while balancing throughput against infrastructure spend.
- Extended reliable call handling from ~2-minute calls to 10–20 minute conversations that stay on script and meet their objectives, and improved average response latency by 10% while keeping the agent accurate and on-track.
- Replaced manual call QA with an automated evaluation module running agent-to-agent calls against adversarial simulated personas (hostile, withholding, erratic), combining signal metrics such as response time, silence and interruptions with LLM judges scoring semantic accuracy and speech fidelity, and pressure-testing inbound and outbound concurrency limits.
- Built out unit and integration coverage, CI/CD pipelines and real-time Grafana monitoring of call performance and system health.
- Built a text chat mode reusing the voice agent's conversational core, letting clients and internal testers regenerate responses, edit history and branch conversations to probe scenarios, removing the need for compute-heavy voice runs during iteration.
- Built a voice audition cleaning pipeline combining audio signal analysis with LLM judges: speed, pitch and tone filtering, stutter and long-gap detection and silence normalisation, filtering at the utterance level and reassembling, producing markedly more stable cloned voices.
- Developed an analysis pipeline that turns call data and per-party sentiment trajectories into scored script recommendations, graded against each script's explicit objectives.
- Developed agentic AI development workflows and custom MCP tooling that dramatically accelerated development time and enabled rapid prototyping, under heavy human oversight with disciplined review to keep code quality high.

Automation Software Engineer: Downer Group

May 2023 – Dec 2025

- Automated data extraction from spreadsheets using Azure Function Apps, stored in a database with a Power BI connection, saving ~\$175k a year per business-unit implementation and eliminating manual transcription errors.
- Built a custom internal Azure dashboard for monitoring project resources across subscriptions, and used it to enact a centralised tagging policy scheduling startup and shutdown events, reducing Azure operating costs by ~60%.
- Automated Konect API queries for critical field work, notifying the requisite authorities to consistently meet SLAs and avoid total penalties of ~\$450k.
- Enhanced security through daily storage account key rotation and SAS generation via Azure Key Vault, reducing 80–100 points of failure to a single centralised point.
- Wrote project setup shell scripts that deploy environments and resources in Azure, linked to custom GitHub branch environments with rules and protections to enable consistent IaC deployments using Bicep.

- Implemented bespoke deep learning and computer vision solutions for real-time artefact recognition with ~94% accuracy.

Engineering Intern: Incat Crowther

Feb 2023 – May 2023

- Produced accurate technical drawings with AutoCAD LT, focusing on double-hull commercial vessels of 30–120 ft.
- Created and modified technical frame and construction drawings for 3 vessels, quickly learning how boats are constructed and outfitted for service.
- Complied with relevant classing authority engineering standards so vessels were safe to operate under varying oceanic conditions.
- Produced precise cut-part, engine and rudder modification, and machinery arrangement drawings for clients within a dynamic environment.

Education

Bachelor of Engineering (Honours) / Computer Science (AI), UNSW Sydney

2019 – 2023

- Honours WAM: 84 (13 High Distinctions, 7 Distinctions).
- Notable marks: MTRN2500 Computing for Mechatronics = 98, COMP3311 Database Systems = 97, COMP3121 Algorithms and Programming Techniques = 96, MMAN3200 Linear Systems and Control = 95.

Higher School Certificate, St Augustine's College Sydney

2018

- ATAR 97.30. Aggregate Cup Recipient (2018), First XI Football team (2018), Stan Arneil Memorial Award Recipient (2017).

Projects

Gym Junkie

- Free, powerful workout tool and data tracker. React Native frontend (Expo) with a FastAPI (Python) backend on Postgres and Redis.
- Live on the Google Play Store and Apple App Store (Android and iOS).
- Lightweight frontend with a focus on increased functionality over existing paid apps.
- Enables users to track their workout progression over time with graph displays.
- Leaderboards, exercise history and muscle maps help users understand their progression.

Other

- Finska: lightweight game tracker for the popular game.
- DownerHelper: PyPI package, a collection of wrappers to eliminate code debt across projects.
- Cell Tracking: segmented cells in a sequence and tracked their positions over time.
- Postgres Deploy: deploys and mirrors DB schemas from a JSON configuration.

Technical Skills

AI / LLMs: Conversational voice agents; LLM fine-tuning and prompt engineering across frontier and open-source models; OpenAI, Anthropic and Groq; speech-to-text (Deepgram) and text-to-speech (Cartesia); RAG with vector search; custom MCP servers; LLM evaluation pipelines and automated testing at scale.

Azure: Function Apps (standard & durable, Python), Container Registry & Instances, Managed Identity, Key Vault, Storage Accounts (+ queues), Event Grid, Postgres; IaC with Bicep and GitHub workflows, az CLI; SharePoint and Logic Apps automation; VM deployment and monitoring; containerised Streamlit apps for ML/AI demos.

AWS: ECS, Secrets Manager, Lambda (zip/containerised), NLB, ElastiCache (Redis), RDS Postgres, CloudFormation, aws CLI.

Python: ML/AI with TensorFlow, OpenCV, scikit-learn, pandas, Pillow; FastAPI for RESTful interfaces; psycopg2 and asyncpg.

Telephony / Networking: SIP, RTP and WebRTC signalling; FreeSWITCH; real-time voice media handling.

Web / Mobile: React (TS), React Native (TS), Express, Jotai.

Other: SQL, PL/pgSQL, Bicep, Docker, C/C++, Shell, HTML, CSS, Java, Cloudflare; SolidWorks, AutoCAD LT and Rhino for complex naval mechanical designs, schematics and visual demonstrations.

Extracurricular

Triathlon, mountain biking, bouldering, tennis and gym.

Referees

Andrew D'Ambrosio · Jared Buncome · Mark Volz · Andrew Tuite

Contact details available upon request.